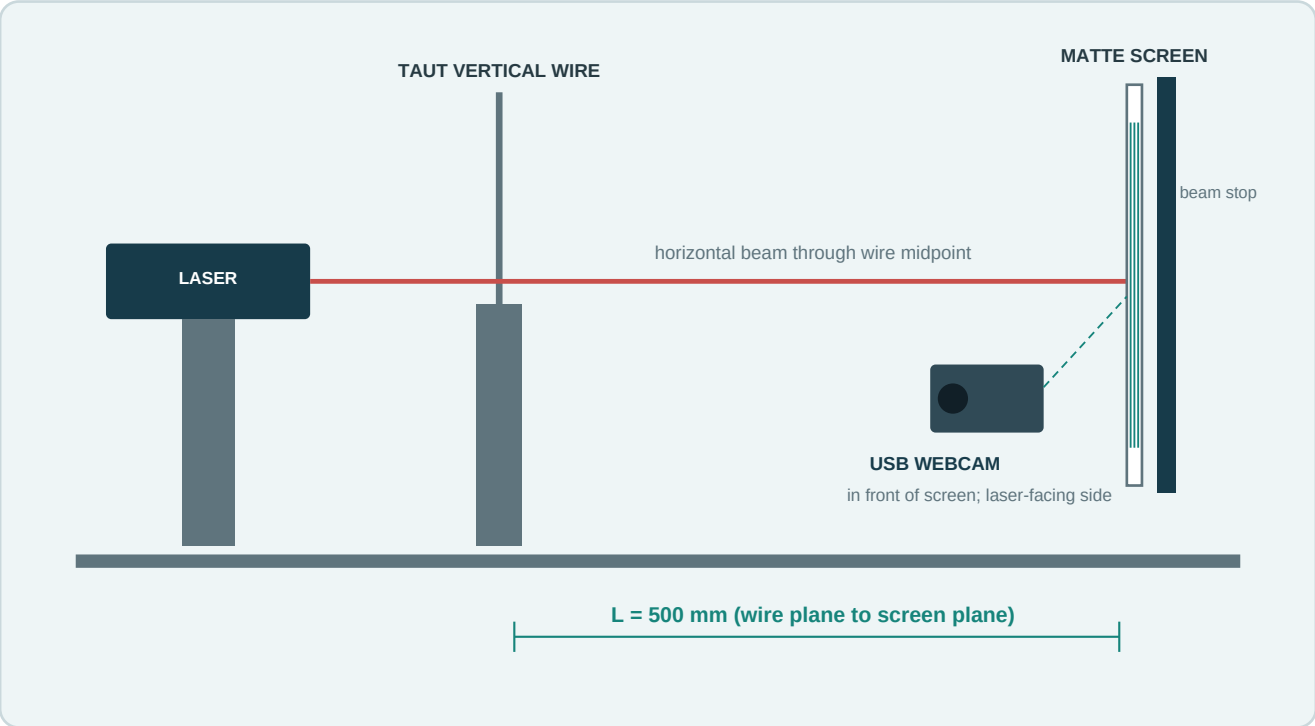
Automatic optical wire micrometer

Current code layout: 500 mm path, USB webcam, 50 mm ruler-click scale calibration

1. Physical arrangement - side view (not to scale)

Laser shines through a vertical taut wire. A matte screen shows the diffraction pattern; the webcam views that screen from the front, on the laser-facing side.



2. Camera and computer signal path

Diffraction
Bright centre and dark bands on the matte card.

USB webcam
Captures screen video; program uses red-channel row average.

Python + OpenCV
Smooths the profile and searches two valleys in the middle 30%.

Live result
Displays estimated diameter, fringe span, camera view and graph.

Image-scale calibration required before a result appears

Place a ruler in the screen plane. Click two visible marks exactly 50 mm apart in the live window. The program converts their pixel separation to mm per pixel. Press R to clear calibration and click again; press Q to quit.

Build, calibrate and interpret

Operating instructions and known limits of the pasted final code

3. Build and alignment sequence

1. **Secure the hardware**
   Make or adapt stable riders for the borrowed bench. Clamp the purchased laser and a simple wire holder. Tension the sample gently; do not kink or break it.
2. **Set the optical path**
   Align the laser horizontally through the vertical wire midpoint. Place the matte screen 500 mm after the wire and perpendicular to the beam. Put the beam stop behind the screen.
3. **Position the webcam**
   Place it about 30 cm in front of the screen on the laser-facing side, below or beside the beam, aimed at the screen. The camera must view the screen, not the direct beam. Focus and frame the ruler marks and fringe pattern.
4. **Reduce glare**
   Dim room light or shade the screen and camera with a black hood. Do not put a filter or cover directly against the webcam lens. Keep the camera and screen still after calibration.
5. **Calibrate**
   Run the Python program. Click two ruler marks that are exactly 50 mm apart and in the same screen plane. Press R to clear and recalibrate if the setup moves.

4. Calculation used by the code

d = 2 x wavelength x wire-to-screen distance / first-minima separation

For 650 nm, L = 500 mm, and d = 127 um: expected minima span is about 5.1 mm.

Use the actual laser wavelength; the current pasted code hard-codes 650 nm and 0.500 m.

5. What the current program actually does

It opens camera index 1 using Windows DirectShow, then retries that same index with OpenCV's default backend if needed. It waits for two ruler clicks 50 mm apart, averages 20 rows from the image centre, uses the red channel, smooths with a Savitzky-Golay filter, and searches for dark valleys in the middle 30% of the image. R resets the scale; Q exits.

Important limitation - inspect the live markers before accepting a number

The pasted algorithm takes the two strongest accepted valleys within its search zone. It does not explicitly ensure one valley is to the left and one to the right of the centre or prove they are the first minima. Screen marks, reflections, or shadows can fool it. Confirm visually that the profile has a bright central region and one clear dark minimum on each side. The code does not save a CSV file.

SAFETY: Use a labelled low-power laser approved by your teacher. Switch it off while adjusting the setup. Never look into the beam or point it at a person or reflective surface. Secure a nonreflective beam stop behind the screen.